ECE 685D HW4

March 4, 2026

Submission Instructions

1. **Upload your Jupyter Notebook (.ipynb file).**
2. **Export all outputs and necessary derivations** as a PDF or HTML (preferred) and upload them.
3. **All mathematical derivations must be typeset in LaTeX.** Handwritten derivations (scanned or photographed) will **not** be accepted.

Use of AI Tools and LLM Policy

The use of large language models (LLMs) and other AI tools is **allowed** and even **encouraged** for this assignment. You may use them to help understand concepts, clarify questions, debug code, or explore ideas.

However:

- You must **clearly disclose how you use AI tools** (e.g., "used ChatGPT to debug training instability").
- You are **fully responsible** for understanding, verifying, and proofreading everything you submit.
- You should be able to explain and justify all derivations and implementation details in your own words.

Learning how to use AI tools effectively and responsibly is an important skill. At the same time, the goal of this assignment is to develop your **own understanding** of the material.

Dataset Reminder

This assignment uses **two different datasets**:

- **Task 1: MNIST**
- **Task 2: Fashion MNIST**

These datasets are **not the same**. Please ensure that you use the correct dataset for each task and clearly indicate which dataset is used in your implementation.

Time Management Reminder

This assignment contains two tasks, each has both theoretical derivations and implementation components, and may require careful debugging and experimentation. **Start early** and avoid leaving the work until the last day before submission.

Task 1. Feature Extraction and Representation Learning (50 pts)

In this problem, we study representation learning from a linear algebraic and neural network perspective. We begin with the classical low-rank approximation problem and Principal Component Analysis (PCA), and then connect these ideas to linear autoencoders. We then extend the discussion to nonlinear autoencoders, and investigate how these models modify the learned representation and reconstruction performance. Let $X \in \mathbb{R}^{d\times N}$ be a centered dataset, and consider the rank-k approximation problem:

$$\min_{M\in\mathbb{R}^{d\times N},\, \mathrm{rank}(M)\le k} \|X - M\|_{\mathrm{F}}^2 .$$

(a) (10pts) Show that if $X = U\Sigma V^\top$ is the singular value decomposition of X, where U, V consist of orthonormal columns, and $\Sigma = \mathrm{diag}\{\sigma_1 \ge \cdots \ge \sigma_{\min(N,d)}\}$, then the optimal solution is

$$M = U_k \Sigma_k V_k^\top ,$$

where U_k and V_k consist of the leftmost k columns of U and V, respectively, and $\Sigma_k = \mathrm{diag}\{\sigma_1 \ge \cdots \ge \sigma_k\}$. Furthermore, show that the minimum error equals

$$\sum_{j=k+1}^{\min(N,d)} \sigma_j^2 .$$

This is also known as the Eckart-Young theorem.

Hint. You can use the following special case of the Hoffman-Wielandt inequality: if $A, B \in \mathbb{R}^{n\times n}$ are symmetric matrices, with eigenvalues $\lambda_1^A \ge \cdots \ge \lambda_n^A$ and $\lambda_1^B \ge \cdots \ge \lambda_n^B$, respectively, then

$$\sum_{j=1}^{n} \left|\lambda_j^A - \lambda_j^B\right|^2 \le \mathrm{tr}\left[(A-B)(A-B)^\top\right] .$$

(b) (10pts) The Principal Component Analysis (PCA) aims to maximize the variance of the projected features:

$$\max_{W\in\mathbb{R}^{d\times k},\, W^\top W = I} \mathrm{Var}(W^\top X) = \mathrm{tr}(W^\top S W),$$

where $S = \frac{1}{N} X X^\top$ is the sample covariance matrix of X. Show that PCA solves

$$\min_{W\in\mathbb{R}^{d\times k},\, W^\top W = I} \left\|X - W W^\top X\right\|_{\mathrm{F}}^2 ,$$

and an optimal W consists of the top k eigenvectors of S.

Linear Autoencoder. Now we consider a linear autoencoder $W_{\mathrm{enc}}, W_{\mathrm{dec}} \in \mathbb{R}^{d\times k}$ with bottleneck k:

$$x \in \mathbb{R}^d, \quad h = W_{\mathrm{enc}}^\top x \in \mathbb{R}^k, \quad \widehat{x} = W_{\mathrm{dec}} h \in \mathbb{R}^d,$$

which is trained by minimizing

$$\mathcal{L}(W_{\mathrm{enc}}, W_{\mathrm{dec}}) = \left\|X - W_{\mathrm{dec}} W_{\mathrm{enc}}^\top X\right\|_{\mathrm{F}}^2 .$$

For simplicity, we assume our autoencoder is underparametrized, i.e. $k < \mathrm{rank}(X)$, so the model cannot perfectly reconstruct X.

(c) (10pts) Show that if $(W_{\text{enc}}, W_{\text{dec}})$ is optimal, then for any invertible matrix $R \in \mathbb{R}^{k\times k}$,

$$\widetilde{W}_{\text{enc}} = W_{\text{enc}} R^\top, \quad \widetilde{W}_{\text{dec}} = W_{\text{dec}} R^{-1}$$

is also optimal. Use (b) to explain that:

- Autoencoder spans the same subspace as PCA, and
- the PCA basis is unique (up to rotation), while encoder/decoder individually are not.

(d) (10pts) Using the MNIST dataset (flattened and centered):

1. Compute PCA (using SVD).
2. Train a linear autoencoder with bottleneck dimension k.
3. Compare:
   - Reconstruction error
   - Principal subspace (e.g., compare projection matrices)
4. Plot: Reconstruction error v.s. bottleneck dimension k.

(e) (Nonlinear autoencoder, 10pts) Replace the linear autoencoder with a nonlinear one:

$$h = f_\theta(x), \quad \widehat{x} = g_\phi(h), \quad x, \widehat{x} \in \mathbb{R}^d,$$

where f_θ and g_ϕ are MLPs, and train this model by minimizing the MSE loss.

Conduct the experiments in (d) on the nonlinear autoencoder, and compare its reconstruction performance with linear autoencoders.

Task 2: From Energy Models to Score-Based Models (50 pts)

In this problem, we study generative modeling from two perspectives:

(1) energy-based models (EBMs), exemplified by Restricted Boltzmann Machines (RBMs), and

(2) score-based models learned via score matching (Hyvärinen, 2005) and denoising score matching.

Throughout, let $p_\theta(x)$ denote a parameterized model density over $x \in \mathbb{R}^d$.

(a) (Gaussian-Bernoulli RBM conditionals, 6pts) A Gaussian-Bernoulli RBM defines the joint distribution over visible $v \in \mathbb{R}^d$ and hidden $h \in \{0,1\}^m$ by

$$p(v,h) \propto e^{-E(v,h)}, \quad \text{where } E(v,h) = -\left(\sum_{i,j} W_{ij} h_j \frac{v_i}{\sigma_i} - \sum_i \frac{(v_i - b_i)^2}{2\sigma_i^2} + \sum_j \alpha_j h_j \right).$$

Derive the conditional distributions:

$$p_\theta(v_i \mid h), \qquad p_\theta(h_j = 1 \mid v).$$

You may leave $p(v_i = x \mid h)$ in integral form once the terms involving h are simplified.

(b) (Implementation: RBM training with CD-k, 10pts) Train a Gaussian-Bernoulli RBM on Fashion MNIST (flattened to dimension $d = 784$) using contrastive divergence (CD-k). Use the standard training/testing split available in PyTorch. Train the model for 25 epochs for hidden dimensions $m \in \{10, 50, 100, 250\}$ where m is the number of hidden units.

1. For each m, report the mean squared reconstruction error (MSE) on the test set. Define the reconstruction of v as one Gibbs step
$$h \sim p(\cdot \mid v), \quad \widetilde{v} \sim p(\cdot \mid h)$$
using the conditional distributions.
2. For one selected value of m (e.g., $m = 100$), plot 16 reconstruction examples from the test set (original image and reconstructed image side-by-side).

(c) (Hyvärinen score matching, 12pts) Let $p_{\text{data}}(x)$ denote the unknown data density on $\mathbb{R}^d$, and define its *score function* as
$$s_{\text{data}}(x) = \nabla_x \log p_{\text{data}}(x).$$
It is known that under mild regularity conditions, a probability density is uniquely determined (up to normalization) by its score function. Therefore, instead of matching densities directly, one may attempt to match their score functions. Let $s_\theta(x)$ be a parametric vector field intended to approximate $s_{\text{data}}(x)$. We consider the score matching objective
$$J(\theta) = \frac{1}{2}\mathbb{E}_{X\sim p_{\text{data}}}\left[\|s_\theta(X) - s_{\text{data}}(X)\|_2^2\right].$$
Assume that the density functions are sufficiently smooth and decay fast at infinity, so that boundary terms vanish and integration by parts is valid. We are going to remove the dependence on s_{data} in the objective function.

1. Show that $J(\theta)$ can be written in the form
$$J(\theta) = \mathbb{E}_{X\sim p_{\text{data}}}\left[\frac{1}{2}\|s_\theta(X)\|_2^2 + \nabla \cdot s_\theta(X)\right] + C,$$
where C is a constant independent of θ. (**Hint:** use integration by parts.)
2. Now suppose $p_\theta(x)$ is an energy-based model
$$p_\theta(x) = \frac{e^{-E_\theta(x)}}{Z(\theta)}, \quad \text{where } Z(\theta) = \int_{\mathbb{R}^d} e^{-E_\theta(x)}\, dx,$$
and $s_\theta(x) = \nabla \log p_\theta(x)$. Express your result from part (c1) entirely in terms of the energy function $E_\theta(x)$ and its derivatives, and explain clearly why the resulting objective does not depend explicitly on the partition function $Z(\theta)$.

(d) (Denoising score matching, 10pts) Directly minimizing the score matching objective in (c) may require computing second derivatives of E_θ, which is inconvenient in deep networks. Instead, we consider the denoising formulation below. Let
$$\tilde{X} = X + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \sigma^2 I_d),$$
where $X \sim p_{\text{data}}$. Define the denoising score matching (DSM) loss
$$\mathcal{J}_{\text{DSM}}(\theta) = \mathbb{E}_{X\sim p_{\text{data}},\, \epsilon\sim\mathcal{N}(0,\sigma^2 I)} \left\| s_\theta(X+\epsilon) + \frac{\epsilon}{\sigma^2}\right\|_2^2.$$
where $s_\theta : \mathbb{R}^d \to \mathbb{R}^d$ is a parametric score network. Show that minimizing $\mathcal{J}_{\text{DSM}}(\theta)$ is equivalent (up to an additive constant and under mild regularity conditions) to minimizing
$$\mathbb{E}_{\tilde{X}\sim p_\sigma} \left\| s_\theta(\tilde{X}) - \nabla \log p_\sigma(\tilde{X})\right\|_2^2,$$
where $\widetilde{X} = X + \epsilon \sim p_\sigma = p_{\text{data}} * \mathcal{N}(0, \sigma^2 I)$ is the Gaussian-smoothed data density.

Hint: Condition the DSM loss on $\widetilde{X}$, and differentiate under the integral expression of p_σ to show

$$\nabla \log p_\sigma(\widetilde{x}) = \mathbb{E}\left[-\frac{\epsilon}{\sigma^2} \,\middle|\, \widetilde{X} = \widetilde{x} \right].$$

(e) (Implementation: score network and Langevin sampling, 12 pts) If $s_\theta(x) \approx \nabla_x \log p_\sigma(x)$, then $s_\theta(x)$ points in the direction of increasing log-density. This suggests the following iterative procedure for generating samples:

$$x_{t+1} = x_t + \eta\, s_\theta(x_t) + \sqrt{2\eta}\,\xi_t, \qquad \xi_t \sim \mathcal{N}(0, I).$$

This update can be interpreted as performing gradient ascent on the log-density, combined with Gaussian noise injection. It is known as (unadjusted) *Langevin dynamics*. In this part, you will train a score network using denoising score matching and generate samples via Langevin dynamics.

1. Implement a neural network

$$s_\theta : \mathbb{R}^{784} \to \mathbb{R}^{784}$$

   (e.g., an MLP) and train this on Fashion MNIST (flattened to dimension 784) using the DSM objective in (d). Choose a noise level $\sigma \in [0.1, 0.5]$. Plot the training loss curve.
2. Initialize $x_0 \sim \mathcal{N}(0, I)$ and run the above iteration for a fixed number of steps. Visualize 16 generated samples. Compare qualitatively with RBM samples from part (b). Discuss how the step size η and the number of steps affect stability and sample quality.

References

A. Hyvärinen. Estimation of non-normalized statistical models by score matching. Journal of Machine Learning Research, 6(Apr):695–709, 2005.